Global Earthquake Risk Analytics using R

Case Study Report

C Chaitanya Kumar Reddy (Roll No. 160124771035)

B.E. Artificial Intelligence and Data Science, Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad

1 Introduction and problem

Earthquakes are among the most damaging natural hazards. How much damage an event can cause depends mainly on its **magnitude** (energy released), its **depth** (shallow events shake the surface harder) and its **location**. A raw catalogue of tens of thousands of events is hard to read, so this case study turns a public catalogue of significant earthquakes into a clear picture of global seismic activity and ranks events by a simple, explainable severity score. The score and models describe *recorded* events; they are not a forecast and ignore population, buildings and local geology.

2 Objectives

1. Clean the catalogue and describe the distribution of magnitude and depth, and the temporal and geographic patterns of earthquakes.
2. Build a transparent 0–100 **risk score** and four risk levels (Low, Moderate, High, Severe), and group earthquakes with K-means clustering.
3. **Predict** whether an event is High/Severe from location, time and catalogue information, and compare three models.

3 Dataset

Source: *Significant Earthquakes, 1965–2016* (USGS earthquake database, distributed on Kaggle: kaggle.com/datasets/usgs/earthquake-database; original catalogue at earthquake.usgs.gov). The file is in the project repository (github.com/chaithanyakumarreddy57-spec/Global-Earthquake-Risk-Analytics) as `data/raw/database.csv`.

Size: 23,412 rows and 21 columns, covering earthquakes of magnitude 5.5 to 9.1 from 1965 to 2016. **Important variables:** `Date`, `Time`, `Latitude`, `Longitude`, `Depth` (km), `Magnitude`, `Magnitude Type`, `Type` (earthquake, nuclear explosion, explosion, rock burst) and `Status` (automatic or reviewed).

Why it is suitable: it is a real, global, well-documented catalogue with exactly the variables a severity analysis needs (where, when, how large, how deep), enough records for clustering and machine learning, and mostly complete core columns (no missing values in date, location, depth or magnitude).

4 Data preparation

The code below runs when the report is knitted. It follows the cleaning steps of `earthquake_analysis.R` (Phases 3, 4 and 7) and ends with a check against the saved clean dataset.

```
eq <- raw
# snake_case names (the full script uses janitor::clean_names(), same result)
names(eq) <- tolower(gsub("[^A-Za-z0-9]+", "_", names(eq)))

# 1. Duplicates: none found, distinct() is a safeguard
n_dup <- sum(duplicated(eq));  eq <- distinct(eq)

# 2. Missing values: drop columns with > 60% missing, then fill the rest
miss_pct <- colMeans(is.na(eq)) * 100
```

```
dropped  <- names(eq)[miss_pct > 60]
eq <- eq[, miss_pct <= 60]
eq$magnitude_type[is.na(eq$magnitude_type)] <- "Unknown"
eq$root_mean_square[is.na(eq$root_mean_square)] <- median(eq$root_mean_square, na.rm = TRUE)

# 3. Incorrect values: keep natural earthquakes only, depth >= 0, valid coordinates
n_before <- nrow(eq)
eq <- eq %>%
  filter(type == "Earthquake") %>%
  mutate(depth = pmax(depth, 0)) %>%                    # 3 depths were slightly negative
  filter(between(latitude, -90, 90), between(longitude, -180, 180))

# 4. Data types: 3 rows hold ISO timestamps, the rest are MM/DD/YYYY + HH:MM:SS
iso <- str_detect(eq$date, "T")
eq$datetime <- as.POSIXct(NA, tz = "UTC")
eq$datetime[iso]  <- ymd_hms(eq$date[iso], tz = "UTC")
eq$datetime[!iso] <- mdy_hms(paste(eq$date[!iso], eq$time[!iso]), tz = "UTC")

# 5. Transformations: time parts and categories
eq <- eq %>% mutate(
  year = year(datetime), month = month(datetime), hour = hour(datetime),
  depth_category = cut(depth, c(-Inf, 70, 300, Inf), right = FALSE,
                       labels = c("Shallow", "Intermediate", "Deep")))

# 6. Risk score (0-100): 60% magnitude energy + 40% shallowness (depth capped at 300 km)
mag_norm   <- (eq$magnitude - 5.5) / (max(eq$magnitude) - 5.5)
depth_norm <- 1 - pmin(eq$depth, 300) / 300
eq$risk_score <- round(100 * (0.6 * mag_norm + 0.4 * depth_norm), 2)
eq$risk_level <- cut(eq$risk_score, c(-Inf, 30, 42, 55, Inf),
                     labels = c("Low", "Moderate", "High", "Severe"))
```

Result. 0 duplicate rows were found; 7 columns with more than 60% missing values were dropped (depth_error, depth_seismic_stations, magnitude_error, magnitude_seismic_stations, azimuthal_gap, horizontal_distance, horizontal_error); 180 non-earthquake records (4 explosion, 175 nuclear explosion, 1 rock burst) were removed. The cleaned dataset has **23,232 earthquakes**. The check above confirms that the rows and risk scores computed here are identical to the saved file `data/processed/earthquakes_clean.rds` used by every other part of the project.

5 Data analysis and visualization

5.1 Magnitude, depth and time

```
p1 <- ggplot(eq, aes(magnitude)) +
  geom_histogram(binwidth = 0.1, fill = "firebrick", colour = "white") +
  labs(title = "Magnitude distribution", x = "Magnitude", y = "Number of earthquakes")
p2 <- ggplot(eq, aes(depth)) +
  geom_histogram(bins = 60, fill = "steelblue", colour = "white") +
  labs(title = "Depth distribution", x = "Depth (km)", y = "Number of earthquakes")
p1 + p2 & theme_minimal(base_size = 9)
```

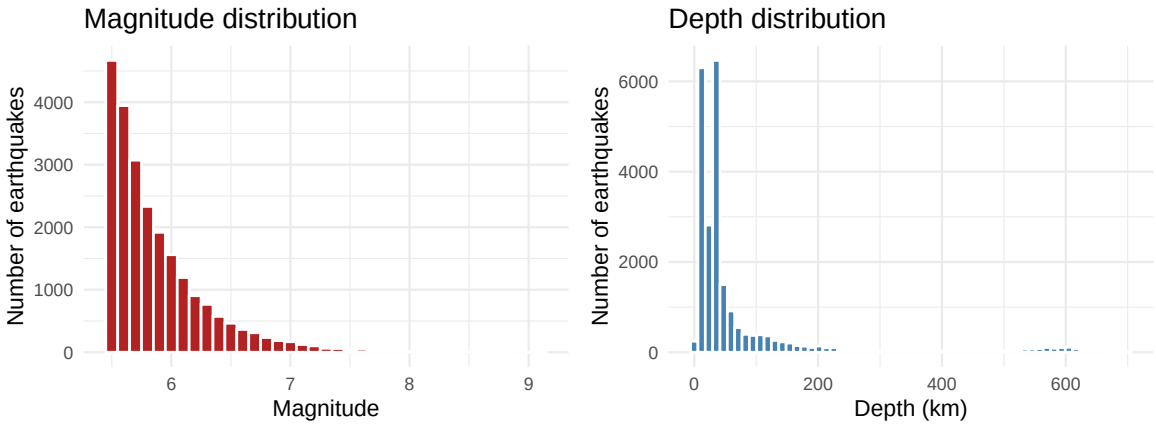


Figure 1: Distribution of magnitude (left) and depth (right).

Magnitudes are strongly right-skewed: the median is 5.70 and only 3.2% of events reach magnitude 7 or more (40 reach 8 or more). Most earthquakes are shallow: 79.5% occur above 70 km (median depth 33 km), and a small group lies deeper than 300 km. Magnitude and depth are almost uncorrelated (Spearman ρ = -0.037), so both are needed in the risk score.

```
yearly <- count(eq, year)
ggplot(yearly, aes(year, n)) +
  geom_line(colour = "firebrick") + geom_point(size = 1) +
  labs(title = "Earthquakes per year", x = "Year", y = "Number of earthquakes") +
  theme_minimal(base_size = 9)
```

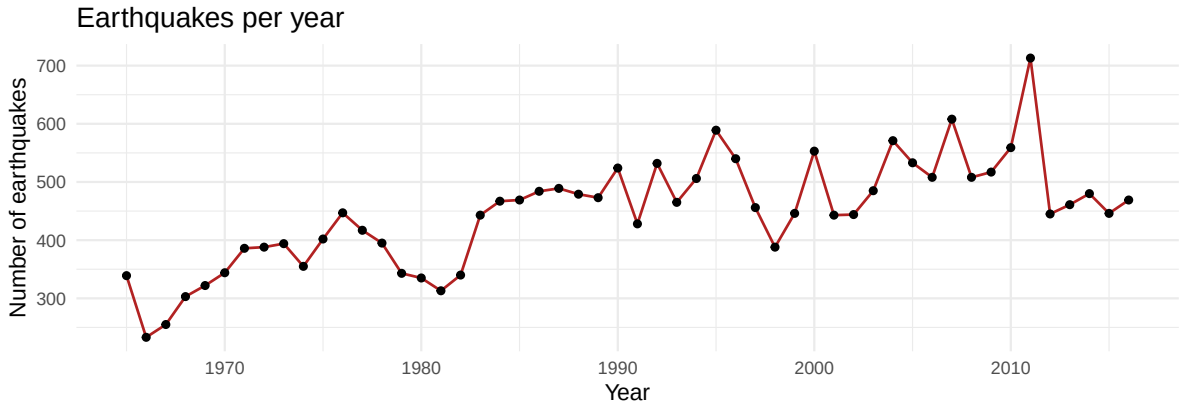


Figure 2: Number of earthquakes per year, 1965-2016.

The yearly count rises by about 4.40 events per year ($p < 0.001$). This is probably influenced by improving detection and reporting rather than a real rise in activity. The busiest year is 2011 (713 events), when the magnitude 9.1 Tohoku earthquake in Japan and its aftershocks occurred.

5.2 Geographic distribution and risk levels

Table 1: Earthquakes by risk level (Low: score up to 30; Moderate: to 42; High: to 55; Severe: above 55).

Risk level	Earthquakes	Share	Mean magnitude	Mean depth (km)
Low	3,375	14.5%	5.78	299.0
Moderate	11,877	51.1%	5.66	37.8
High	6,604	28.4%	6.11	25.0
Severe	1,376	5.9%	6.99	24.6

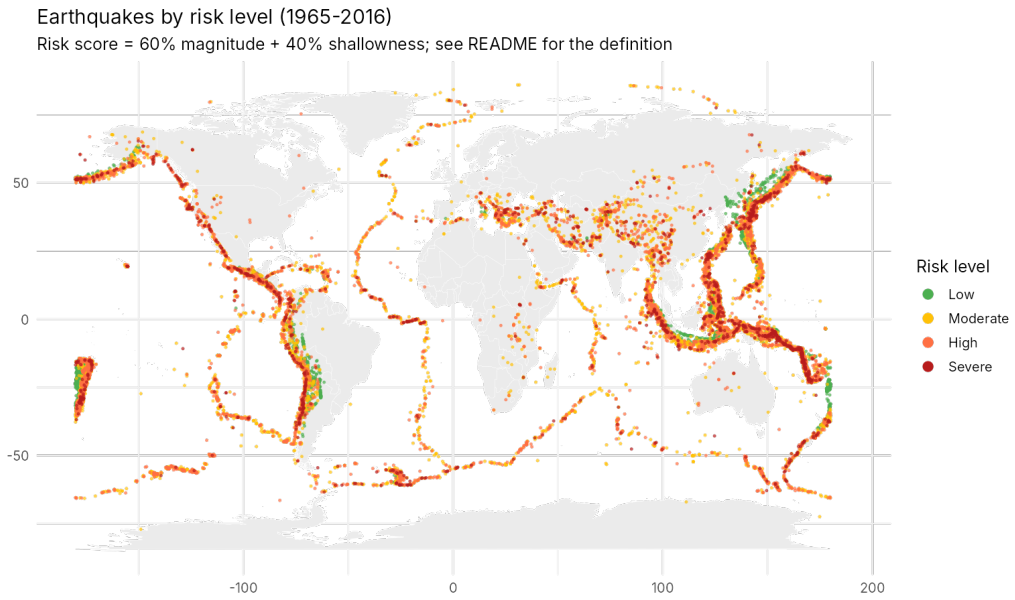


Figure 3: Epicentres coloured by risk level (Severe events drawn on top).

Earthquakes follow the plate boundaries: the Pacific "Ring of Fire", the Alpine–Himalayan belt and the mid-ocean ridges. Moderate is the largest level (51.1%); 34.3% of events are High or Severe, and these concentrate along the subduction zones of the western Pacific and the South American coast. Low-risk events are all deeper than 70 km, because shallowness is part of the score. The most active 10° x 10° cell (20°S–10°S, 160°E–170°E, near Vanuatu) holds 1,307 events.

5.3 Clustering

Table 2: K-means cluster profile (k = 6).

Cluster	Earthquakes	Mean magnitude	Mean depth (km)	Mean risk score
1	8,186	5.67	41.2	37.3
2	2,368	5.75	64.5	35.6
3	3,796	6.28	25.4	49.6
4	1,308	7.10	86.4	57.3
5	2,795	5.83	324.6	15.5
6	4,779	5.70	10.4	41.9

K-means (`nstart = 25`) was run on latitude and longitude converted to 3-D coordinates on the globe (so longitude wraps correctly), log depth and magnitude, all standardised. The clusters differ mainly by depth and magnitude: cluster 5 holds the deep events (mean depth 325 km, lowest risk), cluster 4 the largest events (mean magnitude 7.10, highest risk) and cluster 6 the shallowest. **Caution:** the elbow plot has no sharp bend and the silhouette width is highest at k = 2 (about 0.32, against 0.25 at k = 6), so k = 6 is a descriptive choice, not a strongly supported one.

6 Prediction model

Task. Classify an event as **High/Severe** (the positive class, 34.3% of events) or Low/Moderate. **Why these methods.** Logistic regression is a simple, interpretable baseline; a decision tree gives readable rules; a random forest combines many trees and can capture non-linear interactions between location and time, so it is a natural candidate when location drives the pattern. All three are compared on the same held-out test set. Magnitude and depth are **excluded** from the inputs because they define the risk score (using them would be data leakage); the inputs are latitude, longitude, year, month, hour, review status and magnitude type.

```r
# Abbreviated from earthquake_analysis.R (Phases 8B and 9); not re-run here so that the
# results below are exactly the ones saved in outputs/tables/model_comparison.csv
model_df <- eq %>% transmute(
  high_risk = factor(ifelse(risk_level %in% c("High", "Severe"), "High", "NotHigh"),
                     levels = c("High", "NotHigh")),
  latitude, longitude, year, month, hour, status_code, magnitude_type)
idx   <- caret::createDataPartition(model_df$high_risk, p = 0.8, list = FALSE)  # stratified 80/20
train <- model_df[idx, ];  test <- model_df[-idx, ]
glm_fit  <- glm(high_risk ~ ., data = train, family = binomial)
tree_fit <- rpart::rpart(high_risk ~ ., data = train, method = "class", cp = 0.005)
rf_fit   <- randomForest::randomForest(high_risk ~ ., data = train, ntree = 300, importance = TRUE)
# accuracy, precision, recall, F1: caret::confusionMatrix();  AUC and ROC: pROC::roc()
```

Table 3: Test-set results (positive class = High/Severe; about 4,650 test earthquakes).

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.664	0.561	0.106	0.178	0.647
Decision Tree	0.660	0.526	0.112	0.185	0.616
Random Forest	0.692	0.592	0.336	0.428	0.722

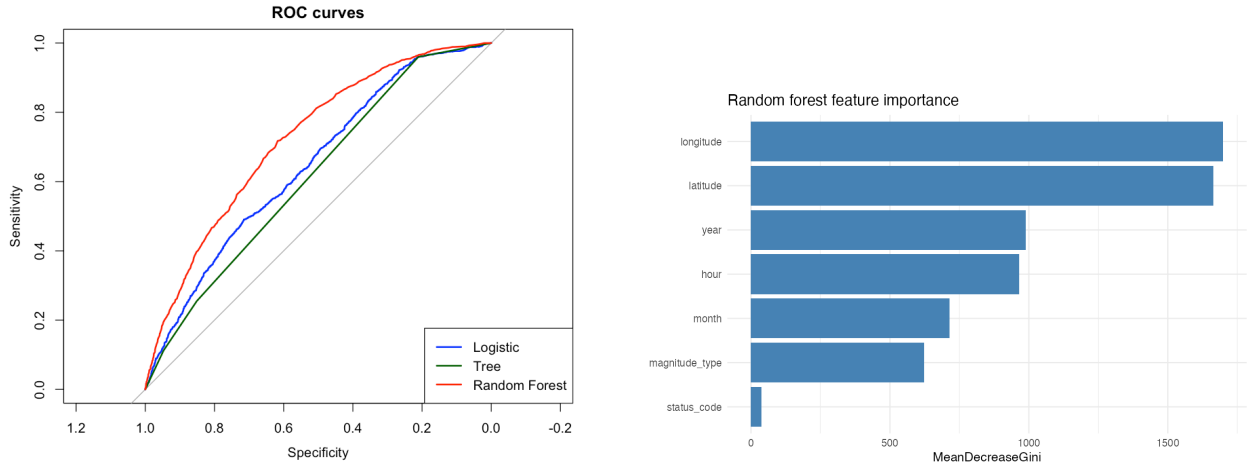


Figure 4: ROC curves of the three models (left) and random forest feature importance (right).

7 Results

The **random forest is the best model on every measure**: accuracy 0.692, precision 0.592, recall 0.336, F1 0.428 and AUC 0.722. Logistic regression and the decision tree barely beat a model that always predicts "not High/Severe" (accuracy 0.657) and find only about 11% of High/Severe events (recall 0.11 and 0.11); the random forest finds about 34%. Its ROC curve lies above the others (AUC 0.72), a moderate ability to rank High/Severe events higher than others. Longitude and latitude are the most important inputs, which reflects the concentration of severe events along particular plate boundaries; `year` and `magnitude_type` are also important, but they describe how an event was catalogued and act partly as proxies for magnitude, so the model should not be read as finding physical causes.

8 Conclusion

From 23,412 raw records, 23,232 natural earthquakes were analysed. Most are shallow (79.5% above 70 km) and lie along plate boundaries, and the explainable risk score places 34.3% of them in the High or Severe levels. A random forest recognised these events better than logistic regression or a decision tree (F1 0.43, AUC 0.72), but only moderately better than a trivial baseline.

Limitations. The risk score uses author-chosen weights (60/40) and ignores exposure; the catalogue is limited to magnitude 5.5 and above; k = 6 is weakly supported; and model skill is modest. Random steps (train/test split, clustering) depend on the random-number stream: re-running the pipeline in another R environment reproduced the cleaning, risk-level counts and model ranking, but not the exact metrics, and

the decision tree collapsed to a single node. Cross-validation would give more stable figures. An interactive Shiny dashboard (`app.R`) and an extended 21-page report (`report/earthquake_report_extended.pdf`) are in the repository.

References

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